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October 25, 2026

Prof. [Editor-in-Chief name]
Editor-in-Chief, *IEEE Transactions on Smart Grid*

Subject: Submission of a regular paper — "Day-Ahead Dispatch of a Grid-Connected Photovoltaic–Battery–Thermal System via Dynamic Programming: A Reproducible Benchmark Study"

Dear Prof. [Editor-in-Chief name],

Please find enclosed our manuscript, submitted for consideration as a regular paper in the *IEEE Transactions on Smart Grid*. The manuscript is seven pages in the two-column IEEEtran format, with four figures, three tables, and seventeen references.

The manuscript formulates the day-ahead dispatch of a grid-connected PV–battery–thermal system as a dynamic program over a discretized state space and solves it exactly. Its contribution is deliberately cast as a benchmark rather than a new algorithm. Every input of the 24-hour instance—585 MW peak load, 21.2% PV energy share, 240 MWh/60 MW battery—is printed in the paper; the reference implementation uses only the Python standard library; and the reported optimum is exact for the discretized problem, with the residual discretization artifact bounded below 0.06% by grid refinement.

We believe this serves a specific need of the journal's readership. A growing literature applies heuristic and learning-based methods to dispatch problems with storage; on small deterministic instances, those methods can now be measured against a certified optimum rather than against their own reported values. The case study also quantifies where the storage value comes from (64% curtailment-penalty avoidance and 36% fuel substitution on this instance), locates the saturation of the storage value with capacity, and shows that the optimal policy is invariant to the curtailment penalty over 100–1000 CNY/MWh.

We confirm that this manuscript is original, has not been published elsewhere, and is not under consideration by any other journal. All authors have read and approved the manuscript and declare no conflicts of interest. The complete instance data appear in the paper; the reference implementation and the figure-generation scripts are available to reviewers upon request and will be released publicly upon acceptance.

Thank you for your consideration.

Sincerely,

First A. Author
(on behalf of all authors)

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